EXPERIMENT - 13

13. Perform Affine Transformation on the given image using python and Open CV

Aim

To perform an affine transformation on a given image using Python and OpenCV.

Procedure

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Select three points from the original image.
4. Define three corresponding points for the transformed image.
5. Apply affine transformation using cv2.getAffineTransform() and cv2.warpAffine().
6. Display the original and transformed images.

Program

```
import cv2
import numpy as np

img = cv2.imread("input.jpg")

if img is None:
   print("Image not found")
else:
   rows, cols = img.shape[:2]

   pts1 = np.float32([[50, 50], [200, 50], [50, 200]])
   pts2 = np.float32([[10, 100], [200, 50], [100, 250]])

   matrix = cv2.getAffineTransform(pts1, pts2)

   result = cv2.warpAffine(img, matrix, (cols, rows))

   cv2.imshow("Original Image", img)
   cv2.imshow("Affine Transformed Image", result)

   cv2.waitKey(0)
   cv2.destroyAllWindows()
```

Sample Input

Input image: input.jpg



Output

Original Image



Affine Transformed Image



Result

Thus, affine transformation was successfully performed on the given image using Python and OpenCV.